

Kashika Tolani

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EDUCATION

Vellore Institute of Technology (VIT)

B.Tech in Electronics and Communication Engineering

Vellore, Tamil Nadu, India

2024 – Current

Indian Institute of Technology, Madras (IIT)

BS in Data Science

Chennai, Tamil Nadu, India

2025 – Current

Loreto Convent Intermediate College

Class XII (Senior Secondary)

Lucknow, Uttar Pradesh, India

Completed 2024

TECHNICAL SKILLS

Languages: Java, Python, C/C++, JavaScript, Swift, Verilog, Assembly (8051), R

Web Development: Flask, React, Node.js, HTML5/CSS3, RESTful APIs, Plotly.js

Data & Analysis: Pandas, NumPy, SciPy, Matplotlib, OpenCV, Scikit-image, Trackpy

Core Concepts: Biomedical Imaging, System Architecture, Image Processing, Interaction Design, Digital Signal Processing, Electromagnetics

Tools & Hardware: Git, SQLite, MongoDB, Arduino, ESP32, Figma

AREAS OF INTEREST

Artificial Intelligence & Machine Learning · Computational Biology · Software Engineering · User Experience (UX) · Data Visualization

TECHNICAL PROJECTS

WoundTrack AI: Platform for Quantitative Wound Healing Analysis

2024 – Present

Flask, NumPy, SciPy, OpenCV, Plotly.js, SQLite

- Spearheaded the end-to-end creation of a web app, **analyzing** over **50+** in-vitro videos for automated evaluation.
- Architected a REST API with an SQLite DB to manage **100+** analysis jobs and securely persist results.
- Collaborated with peers to engineer an OpenCV/Scikit-image pipeline, **documenting** technical specifications and reducing manual calculation time by **85%**.
- Implemented a multi-object cell tracking system to **evaluate 5+** key microscopic metrics with high accuracy.
- Constructed a responsive dashboard visualizing data, enabling the simultaneous comparison of up to **3** experimental conditions.

Smart Study Plan Generator | Python, Generative AI APIs

2025 – 2026

- Directed the design of an AI-powered system utilizing **2** external APIs to **investigate** syllabus data and generate personalized study schedules.
- Championed the implementation of scheduling algorithms to **optimize** learning efficiency, saving users an estimated **4+ hours/week** in manual planning.

UI/UX Design Portfolio | Figma, HTML/CSS

2026

- Partnered with peer reviewers to iterate on a custom landing page, effectively **communicating** design rationale across **5+** high-fidelity interactive screens.
- Prototyped application interfaces structured around **3** distinct user flows to reduce cognitive load and improve layout usability.

Embedded Systems Projects | C/C++, Assembly, Arduino, ESP32

2025 – 2026

- Programmed 8051 microcontrollers using over **500+** lines of Assembly for low-level hardware control and resource optimization.
- Coordinated the integration of **4+** IoT sensors using Arduino and ESP32 to collect data and **presented** real-time hardware findings.